

Disproof of conjecture 00000002684: orbit–subvariety hit counts can grow like $n/2$, beating every polynomial in $\log n$

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1 The conjecture and the readings refuted

Statement (English, verbatim). Definition: Dynamical Mordell–Lang: finiteness of intersections of orbits with subvarieties (the Ghioca–Tucker theorem). Conjecture: A quantitative upper bound on intersection density: the count of intersection points of an orbit with a codimension- r subvariety is a degree- r polynomial in $\log n$ along the iterates, with coefficients explicit in the degree vector of the subvariety. (DML intersection count polynomial law)

For a self-map f , a point x_0 , a subvariety V and $N \geq 0$ write

$$\begin{aligned} c(N) &= \#(\{f^n(x_0) : 0 \leq n \leq N\} \cap V) && \text{(intersection points),} \\ t(N) &= \#\{n \leq N : f^n(x_0) \in V\} && \text{(hitting times).} \end{aligned}$$

In all our examples the orbit is injective, so $c(N) = t(N)$ (and Lean proves both statements). We refute the following readings.

- (R1) *Upper-bound reading.* For every $r \geq 1$ there are f, x_0 and a codimension- r subvariety V such that for *every* real polynomial P (any degree, any coefficients) it is false that $c(N) \leq P(\log N)$ for all sufficiently large N . Hence no choice of coefficients, explicit in the degree vector or not, gives the claimed upper bound, either eventually or for all $N \geq 1$.
- (R2) *Equality reading.* “ $c(N) = P(\log N)$ with $\deg P = r$ ” fails for the same example (equality implies the upper bound). It also fails for orbits whose intersection with V is *finite*: there c is eventually constant, and if $c(N) = P(\log N)$ for all large N then P is a constant, so $\deg P = 0 \neq r$.
- (R3) *Finite intersections, coefficients depending only on V .* For every $r \geq 1$ there is one codimension- r subvariety W (degree vector $(1, \dots, 1)$) and maps g_K ($K \geq 1$) such that the orbit of 0 under g_K meets W in exactly K points, and no single real polynomial P satisfies $c_{g_K}(N) \leq P(\log N)$ for all $K \geq 1$ and all $N \geq 1$.

Readings not refuted. (a) If the claim is restricted to orbits meeting V in finitely many points and the coefficients may depend on f and x_0 (not only on V), the upper bound is trivially true: $c(N)$ is bounded by the total number of intersection points, so a large constant (or any polynomial with positive leading coefficient, for large N) bounds it. This case is consistent with the claim and we do not refute it. (b) Our family in (R3) has maps g_K of degree $\max(1, K - 1)$ (the first coordinate has degree $K - 1$ in z , the last coordinate is $z + 1$); since this grows with K , if

the coefficients may also depend on $\deg f$, (R3) does not apply. (c) The formula for the coefficients in terms of the degree vector is not specified in the statement; (R1) beats every polynomial, so the formula does not matter there.

About the Definition line. The Definition names the Dynamical Mordell–Lang setting. In Witness 1 the set of return times is the even numbers, a single arithmetic progression, and V is invariant under f^2 (which is translation by 2 in z). So Witness 1 is an orbit that meets V infinitely often; it is not excluded by the statement’s wording (“intersection density”, “count ...along the iterates”, “upper bound”). Readers who restrict the claim to finite intersections should look at (R2)–(R3) and at the unrefuted case (a).

2 Conventions and definitions

- Affine space $\mathbb{A}_{\mathbb{Q}}^{r+1}$ is $\mathbb{Q}^{\text{Fin}(r+1)}$; coordinate 0 is x_1 , coordinates $1, \dots, r-1$ are $x_2, \dots, x_r$, and the last coordinate is z . We take $r \geq 1$.
- A polynomial self-map is $p \mapsto (F_i(p))_i$ for a tuple F of polynomials in $\mathbb{Q}[x_1, \dots, x_r, z]$ (Lean: `polyMap`). Iterates start at $f^0 = \text{id}$, and $n \in \mathbb{N}$ includes 0.
- `log` is the natural logarithm `Real.log`; we only use it at $N \geq 1$ (or for large N). Mathlib’s junk value `log 0 = 0` plays no role.
- The line $L_c = \{p : p_i = c_i \text{ for all } i \neq z\}$ through c is the zero locus (`MvPolynomial.zeroLocus`) of the ideal generated by the r polynomials $x_i - c_i$, each of total degree 1; so its degree vector is $(1, \dots, 1)$.
- *Subvariety and codimension.* Lean proves that the vanishing ideal $I(L_c)$ is prime (so L_c is an irreducible Zariski-closed subset), that the coordinate ring $\mathbb{Q}[x_1, \dots, x_r, z]/I(L_c)$ has Krull dimension 1, and that $\mathbb{Q}[x_1, \dots, x_r, z]$ has Krull dimension $r+1$. Hence L_c has codimension r . The proof restricts polynomials to the line: $\rho_c : \mathbb{Q}[x, z] \rightarrow \mathbb{Q}[z]$, $x_i \mapsto c_i$, $z \mapsto z$, is surjective with kernel $I(L_c)$ (since a polynomial over the infinite field $\mathbb{Q}$ vanishing on $\mathbb{Q}$ is zero), so the coordinate ring is isomorphic to $\mathbb{Q}[z]$.

3 The proofs

Lemma 1 (`eventually_eval_log_le`). *For every $P \in \mathbb{R}[X]$, $P(\log N) \leq N/4$ for all sufficiently large $N \in \mathbb{N}$.*

Proof. $P(\log x) = \sum_{k \leq \deg P} a_k (\log x)^k$ and each $(\log x)^k = o(x)$ as $x \rightarrow \infty$ (Mathlib `Real.isLittleO_pow_log_id_atTop`); a finite sum of $o(x)$ terms is $o(x)$. Restrict along $\mathbb{N} \rightarrow \mathbb{R}$. $\square$

Witness 1. $f(x_1, x_2, \dots, x_r, z) = (-x_1, x_2, \dots, x_r, z+1)$, $x_0 = (1, 0, \dots, 0; 0)$, $V = L_{x_0} = \{x_1 = 1, x_2 = \dots = x_r = 0\}$.

Theorem 2 (`conjecture2684_false`, `conjecture2684_disproof`). *For $r \geq 1$: $f^n(x_0) = ((-1)^n, 0, \dots, 0; n)$; $f^n(x_0) \in V$ iff n is even; the orbit is injective; $x_0 \in V$ but $f(x_0) \notin V$. Hence $c(N) = t(N) \geq \lfloor N/2 \rfloor + 1 \geq N/2$ (equality holds in the first inequality; Lean proves only the lower bound), and for every $P \in \mathbb{R}[X]$ it is false that $c(N) \leq P(\log N)$ for all large N (and likewise for t).*

Proof. The orbit formula is an induction. In $\mathbb{Q}$, $(-1)^n = 1$ iff n is even. The z -coordinate is n , so the orbit is injective. The even numbers $2k$, $k \leq \lfloor N/2 \rfloor$, are hitting times, so $t(N) \geq \lfloor N/2 \rfloor + 1 \geq N/2$. If also $t(N) \leq P(\log N) \leq N/4$ for large N , then $N/2 \leq N/4$, which is false for $N \geq 1$. $\square$

The corollary `conjecture2684_false_degree` states the degree- r readings: there is no P of degree r with $c(N) \leq P(\log N)$ for all $N \geq 1$, and none with $c(N) = P(\log N)$ for all $N \geq 1$.

Witness 2 (finite intersections). For $K \geq 1$ let $g_K(x_1, \dots, x_r, z) = (\prod_{j=0}^{K-2} (z - j), x_2, \dots, x_r, z + 1)$, start at 0, and let $W = L_0 = \{x_1 = \dots = x_r = 0\}$ (the z -axis, fixed, independent of K).

Theorem 3 (`g_returns`, `finite_eq_reading`, `finite_uniform_bound_false`). *For $r, K \geq 1$: $g_K^n(0) \in W$ iff $n < K$, so the orbit meets W in exactly K points and $c(N) = K$ for $N \geq K - 1$. If $c(N) = P(\log N)$ for all large N then $P = K$ is constant, so $\deg P \neq r$. And no $P \in \mathbb{R}[X]$ satisfies $c_{g_K}(N) \leq P(\log N)$ for all $K \geq 1$, $N \geq 1$.*

Proof. z increases by 1, the coordinates $x_2, \dots, x_r$ stay 0, and the x_1 -coordinate of $g_K^{m+1}(0)$ is $\prod_{j < K-1} (m - j)$, which vanishes iff $m < K - 1$. If $P(\log N) = K$ for all $N \geq N_0$, then $P - K$ has infinitely many roots (log is injective on $(0, \infty)$), so $P = K$ (`Mathlib Polynomial.eq_of_infinite_eval_eq`). For the last claim pick $N \geq 1$ with $P(\log N) \leq N/4$ and take $K = N + 1$: then $c_{g_K}(N) = N + 1 > N/4$. $\square$

Remark 4. Lean works over $\mathbb{Q}$. The same polynomials give the same orbits over any field of characteristic 0 (this remark is not formalized).

4 Lean formalization and axiom audit

The file `lean/Conjecture2684/Basic.lean` (461 lines, only `import Mathlib`) contains:

- objects: `polyMap`, `hitTimes`, `hitPoints`, `line`, `eqns`, `V`, `W`, `F`, `f`, `x0` (`x` with subscript 0), `G`, `g`;
- variety certificates: `zeroLocus_eqns`, `eqns_totalDegree`, `line_isPrime`, `line_dim`, `ambient_dim`;
- Witness 1: `mem_V_iff_even`, `orbit_injective`, `nondegenerate`, `conjecture2684_false`, `conjecture2684_false_degree`, and the headline `conjecture2684_disproof`: for $r \geq 1$ there are F , x and r equations E of total degree 1 whose zero locus V has prime vanishing ideal and coordinate ring of Krull dimension 1 (ambient dimension $r + 1$), the return set is infinite, and for every real polynomial P the count of intersection points is not eventually $\leq P(\log N)$;
- Witness 2: `g_returns`, `g_orbit_injective`, `finite_eq_reading`, `finite_uniform_bound_false`.

lake build succeeds. `#print axioms` for `conjecture2684_disproof`, `conjecture2684_false`, `conjecture2684_false_degree`, `finite_eq_reading`, `finite_uniform_bound_false`, `g_returns` and `nondegenerate` reports only `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.